

Hardware Report: Prototype Current Stabilization Circuit

Chuwei (David) Liao

July 7, 2026

Contents

1	System Overview	2
1.1	Introduction	2
1.2	Hardware Configuration	2
2	Operational Guide	4
2.1	Connections and Pinouts	4
2.1.1	Power Supply Inputs	4
2.1.2	Load and Signal Interfacing	5
2.1.3	As-Built Grounding Architecture	5
2.2	Standard Operating Procedure	5
2.2.1	Phase 1: Diagnostic and Monitoring Setup	5
2.2.2	Phase 2: Logic and Peripheral Power-Up	6
2.2.3	Phase 3: PID Configuration and Tuning	6
2.2.4	Phase 4: Active Operation	6
2.2.5	Phase 5: Safe Shutdown Sequence	7
2.3	Safety and Absolute Maximum Ratings	7
2.3.1	Absolute Maximum Ratings	7
2.3.2	System Transfer Function	7
2.3.3	Operational Hazards and Best Practices	7
3	Circuit Functionality	8
3.1	High-Level Architecture	8
3.2	Power Stage and Current Path	8
3.3	Current Sensing and Feedback Loop	9
3.3.1	Known Prototype Limitation: Unbuffered Gate Drive	9
3.4	Protection Circuitry	9
4	Performance Assessment	11
4.1	Testing Methodology	11
4.2	Transient Step Response Results	11
4.2.1	Theoretical Verification	11
4.3	Future Characterization	12
4.4	As-Built Hardware Modifications	12
5	Bill of Materials (BOM)	14

Chapter 1

System Overview

1.1 Introduction

The circuit's goal is to produce a quick-switching and low ripple current source controlled by an analog voltage input. The desired load is a set of magnetic coils installed on an H-Bridge to provide bidirectional current switching. The role of the current stabilization circuit is to generate a single directional current that is controlled in magnitude with $V_{setpoint}$. The basic topology is a MOSFET acting as a voltage (V_{GS}) controlled resistor with fixed V_{DS} with feedback from an analog PID at the MOSFET Gate.

1.2 Hardware Configuration

The circuit comes in a metal case, housing 4 units of the circuit for X Bias, Y Bias, Z Bias, and for the MOT coil. All required connections are mounted by either BNC for signals (Feedback, Measure, PWM, DISABLE) or by banana plugs for power. The circuit is assembled by a through-hole breadboard with wires soldered on the back of the boards. There are two cooling fans installed on the side. On the walls of the housing are drivers for the IGBT switch in the circuit. Before attempting to use the circuit please read The operational guide.

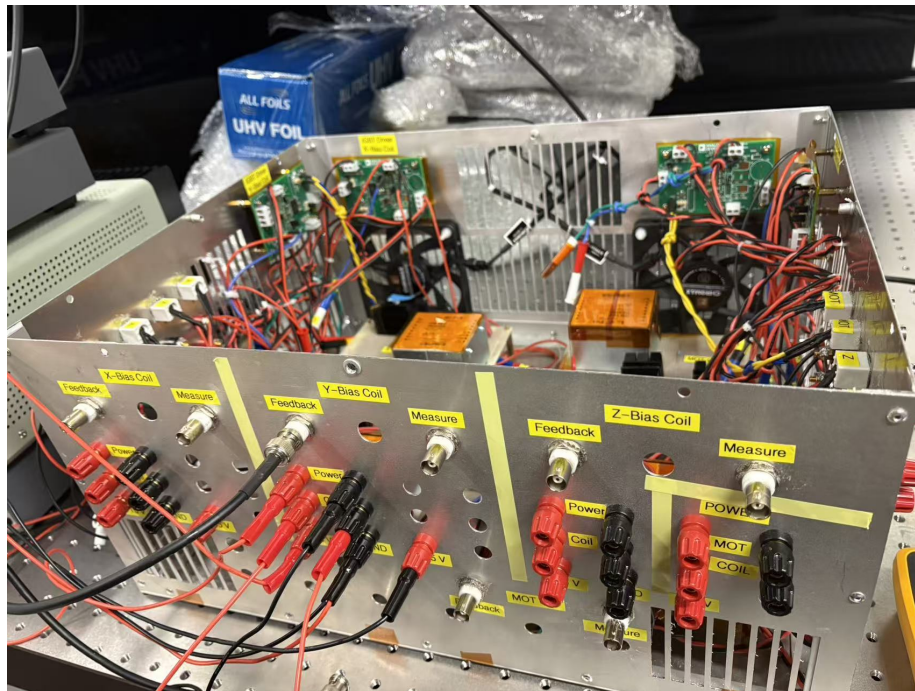


Figure 1.1: Circuit image

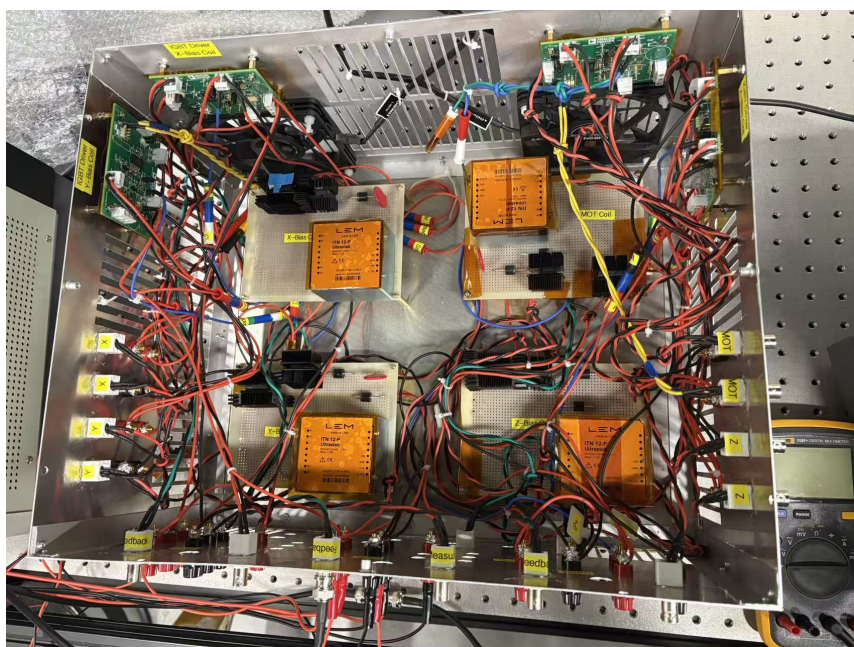


Figure 1.2: Circuit image

Chapter 2

Operational Guide

2.1 Connections and Pinouts

Proper wiring is critical to avoid ground loops, noise injection, or hardware damage. The connections for each axis are categorized into power supplies, load interfacing, control signals, and grounding architecture.

2.1.1 Power Supply Inputs

Main Power (Red/Black): Main drive voltage. Red is positive (+5V to +30V), and Black is the negative return. The applied voltage should be adjusted based on the target current. (*Note: Refer to the Safety and Limits section for maximum current ratings*).

12V Logic (Red/Black): 12V supply for internal circuitry. Red is +12V, Black is GND.

5V Logic (Red/Black): 5V supply for internal circuitry. Red is +5V, Black is GND.

CRITICAL WARNING: Do not swap the 5V and 12V connections or reverse their polarity. Doing so will immediately destroy the IGBT driver.

$\pm 15V$ Bipolar Supply (+15V, GND, -15V): Standard bipolar power for the analog current transducers. *Implementation Note:* If a dedicated bipolar supply is unavailable, this can be synthesized using two isolated single-channel DC supplies. Connect the positive terminal of Supply A to +15V, the negative terminal of Supply B to -15V, and tie the negative of A and positive of B together to form the common GND.

AS-BUILT ROUTING NOTE: Upon physical inspection of the enclosure, the **Z-Bias** and **MOT** axes share a single $\pm 15V$ power supply line for their transducers. Do not attempt to wire redundant or conflicting $\pm 15V$ supplies to these two axes.

2.1.2 Load and Signal Interfacing

Coil (Load): Connects the inductive load to the stabilization circuit. Current flows from the Red terminal to the Black terminal.

PID Measure: Connects directly to the Measure input terminal on the external PID controller.

PID Feedback: Connects directly to the Output/Feedback terminal on the external PID controller.

Setpoint Control: Connects the PID setpoint to your external analog control voltage source (e.g., an NI DAQ system or a function generator). **Note:** This source must be strictly high-precision and low-noise to ensure stable current output.

Disable / PWM: These 5V logic inputs control the IGBT driver state (5V on the PWM pin enables the driver; 5V on the Disable pin turns it off). *Note: In this prototype revision, these states are hardwired internally to be permanently ON and OFF, respectively.*

2.1.3 As-Built Grounding Architecture

Physical inspection of the prototype enclosure reveals specific internal ground ties that must be accounted for when connecting external power supplies and oscilloscopes. Do not assume galvanic isolation between these nodes:

- **Power and Logic Ground Tying:** For both the **X-Bias** and **Z-Bias** axes, the Main Power negative return (Power -) is physically connected to the 12V logic ground (12V -).
- **Measurement Grounds:** Across all axes, the external "Measure" ground terminal (connecting to the PID controller) is internally tied to the internal Current Transducer ground.

CAUTION: Because the main power return is tied to the 12V logic ground on the X and Z axes, you must ensure that your external power supplies do not create an accidental ground loop through earth ground. Ensure your main power supplies have floating outputs before connecting them to this topology.

2.2 Standard Operating Procedure

To safely initialize, tune, and monitor the current stabilization circuit, follow this chronological sequence. Ensure all load and power connections (detailed in the previous section) are secure before beginning.

2.2.1 Phase 1: Diagnostic and Monitoring Setup

Before applying power, establish your diagnostic sightlines to ensure the circuit is behaving as expected. All control and feedback signals in this system are analog voltage levels.

1. **Current Monitoring:** Connect a BNC T-adapter (tee) at the **Measure** terminal to monitor the current transducer's output voltage via an oscilloscope. Alternatively, you may clamp a current probe directly onto the **Coil** output terminal.
2. **Command Monitoring:** Connect a second T-adapter at the **Setpoint** input to monitor your command voltage on the oscilloscope.

2.2.2 Phase 2: Logic and Peripheral Power-Up

1. Energize the logic and analog front-end supplies first: turn on the **12V**, **5V**, and **±15V** supplies.
2. Turn on the external analog PID controller.

2.2.3 Phase 3: PID Configuration and Tuning

Before energizing the main power, the PID controller must be clamped and configured to prevent immediate current spikes.

1. Set the PID output voltage limits to a strict range of **0V (Min)** to **10V (Max)**.
2. Input the baseline tuning parameters. A known-good starting configuration is:
 - **Proportional (P):** 0.8
 - **Integral (I):** 2.6×10^5
 - **Derivative (D):** 0
 - **Offset:** 4.5
3. Fine-tune these parameters further as required for your specific load response.

2.2.4 Phase 4: Active Operation

1. Turn on the enclosure cooling fans.
2. Energize the **Main Power** supply.
3. Begin inputting your analog command voltage into the Setpoint terminal.

CAUTION: If operating the system without active fan cooling, closely monitor the physical hardware. The MOSFETs are highly susceptible to rapid heating and thermal runaway under load.

2.2.5 Phase 5: Safe Shutdown Sequence

To safely power down the axis and prevent inductive kickback or control loop runaway, this sequence must be followed in order:

1. Turn **OFF** the external PID controller to kill the command signal.
2. Turn **OFF** the Main Power supply.
3. Turn **OFF** all peripheral and logic power (12V, 5V, and $\pm 15V$).

2.3 Safety and Absolute Maximum Ratings

Failure to adhere to the following limits may result in catastrophic failure of the active components, thermal runaway, or severe hardware damage.

2.3.1 Absolute Maximum Ratings

- **Maximum Current (Load): 12.5 A.** Do not exceed this value under any circumstances, as it is the hard limit of the internal current transducer.
- **Maximum Voltage (Main Power): 100 V.** Exceeding this limit will risk dielectric breakdown and component failure.
- **Peripheral Logic Voltages:** Strictly adhere to the labeled voltages (5V, 12V, $\pm 15V$). Over-volting the logic or analog rails will destroy the front-end ICs and IGBT drivers.

2.3.2 System Transfer Function

When inputting your analog setpoint voltage, you must strictly calculate your target current using the system's transfer function.

CRITICAL SETPOINT CONVERSION:

1 A output = 0.4 V input command

(Equivalently: The system gain is 2.5 A/V)

2.3.3 Operational Hazards and Best Practices

- **Pre-Flight Enclosure Check:** Before applying any power, verify with a multimeter that there are no dead shorts between the Main Power rails and the physical metal enclosure.
- **Thermal Runaway:** The linear MOSFET stages dissipate significant power as heat. Monitor temperatures closely during operation, especially if operating without forced-air cooling.
- **No Hot-Plugging: NEVER** connect or disconnect the coil, power supplies, or logic headers while the system is energized. Doing so can cause massive inductive voltage spikes or floating-gate states that will destroy the silicon.

Chapter 3

Circuit Functionality

3.1 High-Level Architecture

The core of the stabilization circuit operates as a high-power, linear Voltage-Controlled Current Source (VCCS). Rather than using high-frequency Pulse Width Modulation (PWM) which can introduce switching noise, this topology operates the primary series MOSFET in its linear (active) region. By continuously modulating the MOSFET's gate voltage via an external feedback loop, the circuit acts as a dynamic variable resistor, maintaining a highly stable and precise DC current through the inductive load regardless of thermal drift or load impedance variations.

3.2 Power Stage and Current Path

The primary high-current path flows sequentially through the active control elements and the load:

1. **Control Element (MOSFET):** Main power (V_{power}) enters the high-side MOSFET at the drain terminal. The MOSFET throttles the current

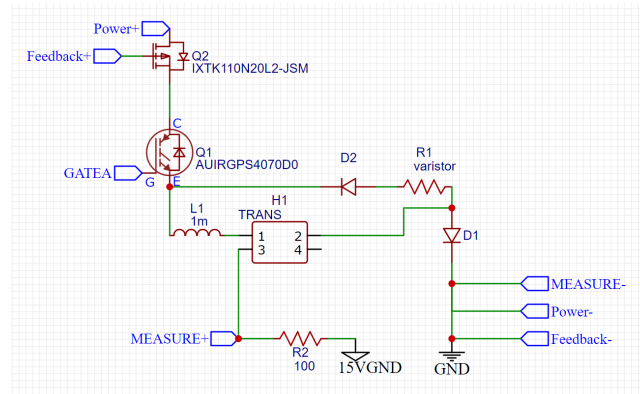


Figure 3.1:

and outputs it from the source terminal.

2. **Static Switch (IGBT):** Current flows from the MOSFET source into the collector of a series IGBT, exiting via the emitter. While the IGBT is designed to allow hard ON/OFF switching, in this operational mode, its gate driver is continuously energized (hardwired ON), acting merely as a static pass-through.
3. **Load and Sensing:** The current passes through the inductive coil load, exits the coil, and flows through the primary aperture of the current transducer before returning to the main power ground.

3.3 Current Sensing and Feedback Loop

To achieve closed-loop stabilization, the actual load current is continuously measured and fed back to the external PID controller.

- **Current Transducer:** A precision LEM ITN 12-P ULTRASLAB current transducer sits in series with the load return path. It possesses a 1:250 secondary winding ratio.
- **Burden Resistor:** The scaled secondary current is driven through a precision $100\ \Omega$ burden resistor to ground. By Ohm's Law, every 1 A of primary load current induces 4 mA in the secondary, generating exactly 0.4 V across the resistor.
- **Gate Control (Unbuffered):** This 0.4 V/A analog signal is sent to the external PID controller's Measure terminal. The PID controller compares this to the Setpoint and adjusts its Output voltage to zero out the error.

3.3.1 Known Prototype Limitation: Unbuffered Gate Drive

In this specific prototype revision, the 0–8V analog control signal from the PID controller is fed *directly* into the gate of the main linear MOSFET (IXTK60N50L2) without an intermediate buffer, op-amp, or dedicated gate driver.

Because the PID controller's output stage must directly charge and discharge the MOSFET's substantial input capacitance (C_{iss}), the system's slew rate and overall control loop bandwidth are heavily restricted. Furthermore, this direct connection leaves the PID output vulnerable to transient charge injection via the MOSFET's Miller capacitance. This is a known flaw that will be addressed in future revisions via an active driver stage.

To charge 610nC of gate charge of MOSFET with 20mA max current of PID, introduces 30us rise lag.

3.4 Protection Circuitry

Because the circuit drives a highly inductive coil, abrupt interruptions in current flow (e.g., emergency shutdowns or power failures) will generate massive transient voltage spikes ($V = L \frac{di}{dt}$). The circuit employs three layers of protection to safely dissipate this flyback energy:

- **Flyback Diode:** Placed in reverse-bias across the series combination of the coil and current transducer. If the primary switch opens, the collapsing magnetic field of the coil will forward-bias the diode, allowing the inductive current to safely loop and dissipate as heat.
- **Varistor (MOV):** Placed in parallel with the flyback diode to clamp high-voltage transients that exceed the diode's reverse breakdown threshold.
- **Ground-Clamping Diode:** A dedicated diode ties the output of the current transducer to ground, preventing the node from swinging to a lethal negative voltage during a transient event, thereby protecting the transducer's internal electronics.

Chapter 4

Performance Assessment

4.1 Testing Methodology

The performance of the prototype stabilization circuit was evaluated primarily on its transient time-domain response. The objective was to quantify the slew rate limitations imposed by the unbuffered MOSFET gate drive. An external square wave was fed into the PID Setpoint terminal to command a step-change in current, and the output was monitored via the current transducer's Measure terminal.

4.2 Transient Step Response Results

A low-frequency square wave was utilized to command a step-response from the stabilization loop. The resulting transition times were recorded:

- **Rise Time (t_r):** 26 μs
- **Fall Time (t_f):** 30 μs

4.2.1 Theoretical Verification

The measured transient times align perfectly with the theoretical limits of the unbuffered gate drive architecture. The IXTK60N50L2 MOSFET possesses a massive total gate charge (Q_g) of approximately 610 nC. The external analog PID controller's output stage is bottlenecked by a maximum source/sink current (I_{max}) of approximately 20 mA.

Using the standard charge equation $t = \frac{Q_g}{I_{max}}$, the theoretical minimum time to fully charge or discharge the gate is:

$$t = \frac{610nC}{20mA} = 30.5\mu s$$

The measured 30 μs fall time and 26 μs rise time reflect this exact physical limitation. Furthermore, these transition speeds approach the approximate 100 kHz operational bandwidth limit of the analog PID controller itself.



Figure 4.1: Rise time image

4.3 Future Characterization

Comprehensive small-signal frequency response analysis, closed-loop phase margin testing, and steady-state thermal drift characterization are explicitly outside the scope of this initial prototype hardware report. These advanced metrics are deferred the original PhD thesis by Mithilesh Kumar Parit.

4.4 As-Built Hardware Modifications

During the commissioning of the 4-axis enclosure, a hardware failure was identified on the **X-Bias** axis.

- **Issue:** The isolated IGBT gate driver (EVAL-ADUM4221-1EBZ) on the X-Bias board is non-functional, leaving the series IGBT stuck in an open (OFF) state.
- **Resolution:** A physical bypass jumper was soldered directly from the main MOSFET Source terminal to the IGBT Emitter terminal. The X-Bias axis operates normally, but the static IGBT switch is entirely bypassed in the current path.

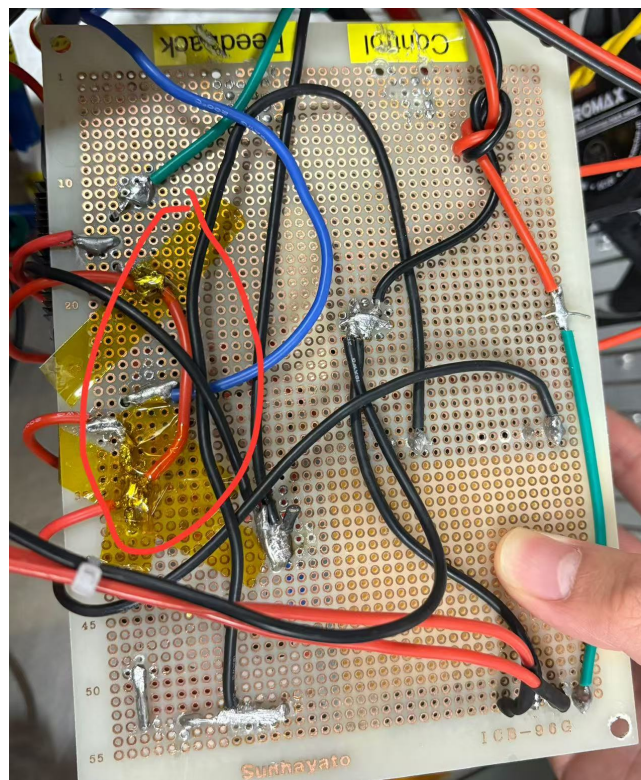


Figure 4.2: Jumper Wire

Chapter 5

Bill of Materials (BOM)

The following table details the primary active components, sensing hardware, and protection circuitry utilized in this prototype revision.

Component	Part Number	Description / Function
Linear MOSFET	IXTK60N50L2	500V, 60A Linear Power MOSFET (Main VCCS control element).
IGBT	AUIRGPS4070D0	400V, 70A Automotive Trench IGBT (Static series switch). <i>Note: Bypassed on X-Bias axis.</i>
IGBT Driver	EVAL-ADUM4221-1EBZ	Isolated half-bridge gate driver evaluation board.
Current Transducer	LEM ITN 12-P ULTRASLAB	High-precision closed-loop transducer (12.5A, 1:250 ratio).
Burden Resistor	100 Ω Precision	Converts the 4mA/A transducer secondary current into the 0.4V/A feedback voltage.
Ref. Resistors	3.3 k Ω	Establishes reference voltage dividers.
TVS Diode	5KP58A	5000W Transient Voltage Suppressor (Clamps inductive spikes).
Varistor (MOV)	V150LA20AP	150V RMS Metal-Oxide Varistor (Parallel overvoltage protection).

Table 5.1: Primary hardware components for a single stabilization axis.